3.5 Graphing Linear Equations in Standard Form

Name _____ Period ____ Date _____

Algebra I 3.5 · Student sheet

I can…

- graph horizontal and vertical lines.
- find the intercepts of Ax + By = C and use them to graph.
- interpret the intercepts in real-life situations.

Vocabulary · fill in the missing word

Standard form — Ax + By = _____, where A and B are not both 0

Horizontal line — y = _____

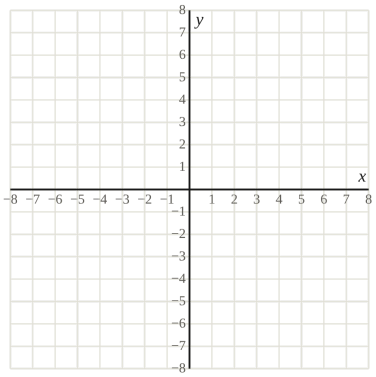
Vertical line — x = a (not a _____)

Intercepts — _____: let y = 0; y-intercept: let x = 0

Part 1 · Snack budget

You have $12 for granola bars ($2 each, x) and fruit cups ($1.50 each, y).

Equation: _____ x-intercept: _____ y-intercept: _____



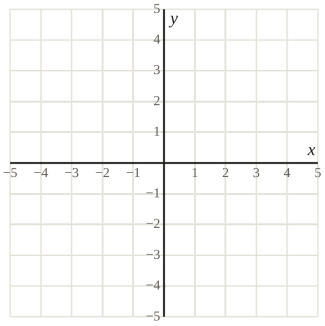
Combinations that use exactly $12: _____

Part 2 · Notes and examples (fill in each note, then work the example)

Horizontal and vertical lines: y = b is a _____ line; x = a is a _____ line.

Example 1

Graph y = −3 and x = 2.

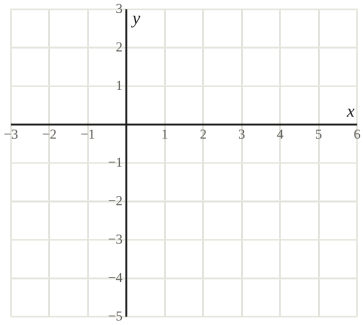


Standard form: Standard form: _____.

Intercepts: x-intercept: let _____ = 0. y-intercept: let _____ = 0.

Example 2

Find the intercepts of $3x - 4y = 12$ and graph.



Example 3

You have $30 to spend on songs ($1 each) and movies ($5 each): $x + 5y = 30$. Find and interpret the intercepts.

Part 3 · Partner practice

Partner 1

Find the intercepts of $-2x + y = 6$.

Partner 2

Graph $x = -4$. Is it a function?